Ch 6.2: Shrinkage - Ridge regression

Lecture 17 - CMSE 381

Michigan State University
::
Dept of Computational Mathematics, Science & Engineering

Wed, Feb 25, 2026

Announcements

Last time:

- Subset selection

This time:

- Ridge regression

Announcements:

- HW #3 due Sunday 3/8

11	F	2/6	Multiple Logistic Regression / Multinomial Logistic Regression	4.3.4-5	HW #2 Due Mon 2/9	
	M	2/9	***Project Day & Review***			
	W	2/11	**Midterm #1**			
12	F	2/13	**Class not held**			
13	M	2/16	Leave one out and k-fold CV	5.1.1-3		
14	W	2/18	More k-fold CV	5.1.4-5		Q5
15	F	2/20	k-fold CV for classification	5.1.5		
16	M	2/23	Subset selection	6.1		
17	W	2/25	Shrinkage: Ridge	6.2.1		
18	F	2/27	Shrinkage: Lasso	6.2.2		
	M	3/2	Spring Break			
	W	3/4	Spring Break			
	F	3/6	Spring Break		HW #3 Due Sun 3/8	
19	M	3/9	PCA	6.3		
20	W	3/11	PCR	6.3		Q6

Section 1

Last time

Subset selection

Algorithm 6.1 *Best subset selection*

1. Let $\mathcal{M}_0$ denote the *null model*, which contains no predictors. This model simply predicts the sample mean for each observation.
2. For $k = 1, 2, \ldots p$:
   (a) Fit all $\binom{p}{k}$ models that contain exactly k predictors.
   (b) Pick the best among these $\binom{p}{k}$ models, and call it $\mathcal{M}_k$. Here *best* is defined as having the smallest RSS, or equivalently largest R^2.
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2.

Algorithm 6.2 *Forward stepwise selection*

1. Let $\mathcal{M}_0$ denote the *null* model, which contains no predictors.
2. For $k = 0, \ldots, p-1$:
   (a) Consider all $p-k$ models that augment the predictors in $\mathcal{M}_k$ with one additional predictor.
   (b) Choose the *best* among these $p-k$ models, and call it $\mathcal{M}_{k+1}$. Here *best* is defined as having smallest RSS or highest R^2.
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2.

Algorithm 6.3 *Backward stepwise selection*

1. Let $\mathcal{M}_p$ denote the *full* model, which contains all p predictors.
2. For $k = p, p-1, \ldots, 1$:
   (a) Consider all k models that contain all but one of the predictors in $\mathcal{M}_k$, for a total of $k-1$ predictors.
   (b) Choose the *best* among these k models, and call it $\mathcal{M}_{k-1}$. Here *best* is defined as having smallest RSS or highest R^2.
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2.

What should you learn from this lecture?

- What is regularization? Why do we need it?
- What are the two basic types of regularization methods? How are they implemented mathematically in linear regression?
- How do you fit a ridge regression model in python?
- How do you control the model flexibility & bias-variance tradeoff when using regularization?
- How do you find the right amount of regularization using cross-validation? How do you do this in python?
- What additional precautions do you need to take when using regularization (compared to least squares)?
- What are the advantages of regularization compared to Least Squares?
- What are the advantages of regularization compared to subset selection?

Section 2

Ridge Regression

Goal

- Fit model using all p predictors
- Aim to constrain (regularize) coefficient estimates
- Shrink the coefficient estimates towards 0

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4$$

- Ridge regression
- Lasso

Ridge regression

Before:

$$RSS = \sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2$$

- $\lambda \geq 0$ is a tuning paramter to figure out separately.
- Second term is called "shrinkage penalty"
- $\lambda = 0$ means least squares estimate
- first encouter of regularization, l_2 norm, l_p norm
- regularization = constrain = less flexibility

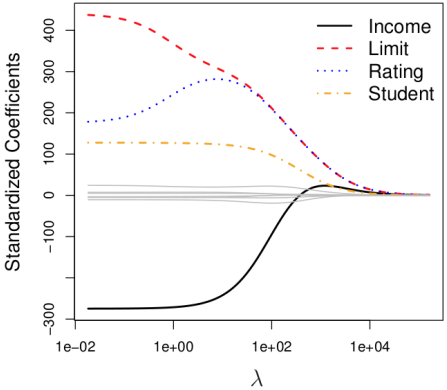
After:

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}\beta_j^2 = RSS + \lambda\sum_{j=1}^{p}\beta_j^2$$

- As $\lambda \to \infty$ impact of shrinkage penalty grows, ridge regression coeff will go towards 0
- Not applied to the intercept
- call coefficients found by ridge regression $\hat{\beta}_\lambda^R = (\beta_{1,\lambda}^R, \cdots, \beta_{p,\lambda}^R \cdots)$

Example from the `Credit` data

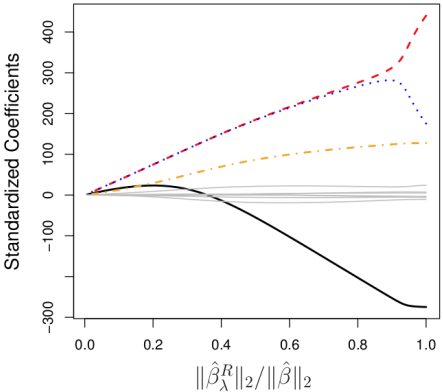
$$RSS + \lambda \sum_{j=1}^{p} \beta_j^2$$



- Left x-axis, $\lambda \approx 0$
- Each line is the predicted coefficient value for a single variable
- as λ increases everything goes to 0, basically giving the null model at the right end

Same Setting, Different Plot

$$RSS + \lambda \sum_{j=1}^{p} \beta_j^2 \qquad \|\beta\|_2 = \sqrt{\sum_{j=1}^{p} \beta_j^2}$$



- $\hat{\beta}$ is the vector of least squares coeff estimates
- $\|\cdot\|$ measures distance from 0
- $\|\hat{\beta}_\lambda^R\|/\|\hat{\beta}\|$ ranges from 1 (when $\lambda = 0$) down to 0
- Amount the coefficient estimates have been shrunk
- more comparable across different regularization methods
- left more flexible, right less flexible

Scale equivavariance (or lack thereof)

Scale equivariant: Multiplying a variable by c (cX_i) just returns a coefficient multiplied by $1/c$ ($1/c\beta_i$)

- Ex: `income variable`.
- Least squares is scale equivariant
- Ridge regression very much is not
- $X_j\hat{\beta}^R_{j,\lambda}$ depends not only on λ but also on values of other predictors

Solution: Standardize predictors

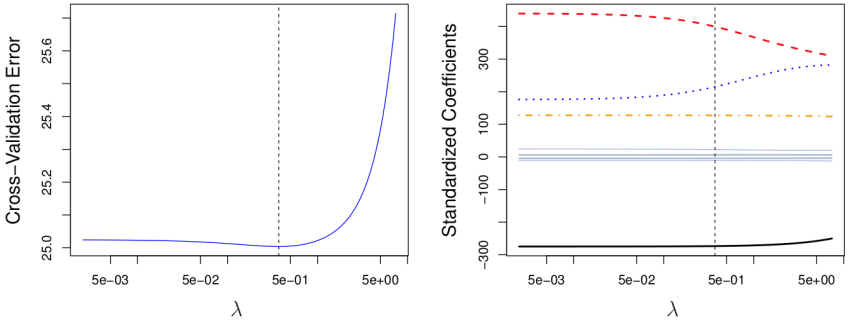
$$\widetilde{x}_{ij} = \frac{x_{ij}}{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \overline{x}_j)^2}}$$

- Denominator is estimated standard deviation of the jth predictor
- Standardized predictors will all have a standard deviation of one
- Previous figures showe standardized ridge regression coeffs on the y-axis

Using Cross-Validation to find λ

- Choose a grid of λ values
- Compute the (k-fold) cross-validation error for each value of λ
- Select the tuning parameter value λ for which the CV error is smallest.
- The model is re-fit using all of the available observations and the selected value of the tuning parameter.

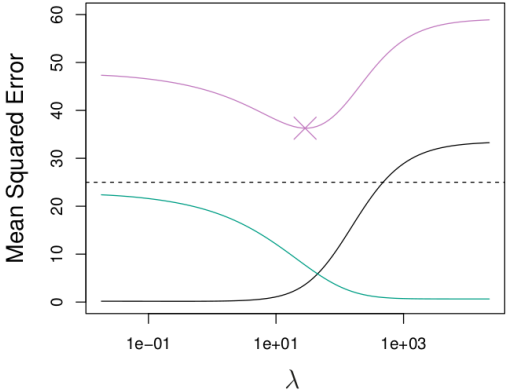
LOOCV choice of λ for ridge regression and `Credit` data



- Dashed vertical lines indicate the selected value of λ.
- In this case, small λ so optimal fit involves small amount of shrinkage relative to least squares
- The dip is not very pronounced, so there is rather a wide range of values that would give a very similar error.
- In a case like this we might simply use the least squares solution.

Coding

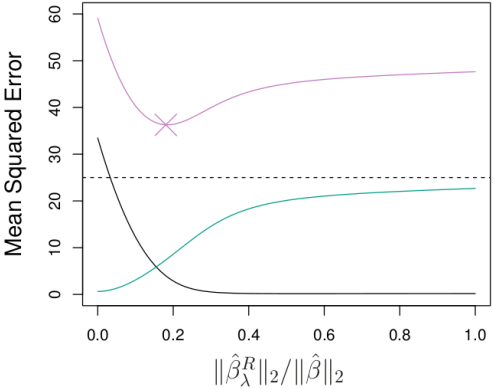
Bias-Variance tradeoff



Squared bias (black), variance (green), and test mean squared error (purple) for simulated data. *Horizontal dashed line is minimum possible test MSE*

- Bias-variance tradeoff
- Minimum MSE achieved around $\lambda = 30$
- High variance means MSE for $\lambda = 0$ (least squares) and $\lambda = \infty$ (null model) are basically the same

More Bias-Variance Tradeoff



Squared bias (black), variance (green), and test mean squared error (purple) for simulated data.

Advantages of Ridge

Ridge vs. Least Squares:

- Previous slide and ability to lower variance
- Ridge regression works best in situations where the least squares estimates have high variance
- Can trade off small increase in bias for a large decrease in variance

Ridge vs. Subset Selection:

- So much better computationally!
- Only fits a single model rather than 2^p
- Tricks available to solve all for all values of λ simultaneous

Look back and look ahead

11	F	2/6	Multiple Logistic Regression / Multinomial Logistic Regression	4.3.4-5	HW #2 Due Mon 2/9	
	M	2/9	***Project Day & Review***			
	W	2/11	**Midterm #1**			
12	F	2/13	**Class not held**			
13	M	2/16	Leave one out and k-fold CV	5.1.1-3		
14	W	2/18	More k-fold CV	5.1.4-5		Q5
15	F	2/20	k-fold CV for classification	5.1.5		
16	M	2/23	Subset selection	6.1		
17	W	2/25	Shrinkage: Ridge	6.2.1		
18	F	2/27	Shrinkage: Lasso	6.2.2		
	M	3/2	Spring Break			
	W	3/4	Spring Break			
	F	3/6	Spring Break		HW #3 Due Sun 3/8	
19	M	3/9	PCA	6.3		
20	W	3/11	PCR	6.3		Q6

- What is regularization? Why do we need it?
- What are the two basic types of regularization methods? How are they implemented mathematically in linear regression?
- How do you fit a ridge regression model in python?
- How do you control the model flexibility & bias-variance tradeoff when using regularization?
- How do you find the right amount of regularization using cross-validation? How do you do this in python?
- What additional precautions do you need to take when using regularization (compared to least squares)?
- What are the advantages of regularization compared to Least Squares?
- What are the advantages of regularization compared to subset selection?